Ticket 4

TCK04-AccessReq

Summary

Category	Access management
Requester	Bob Nasser (Sales)
Request	Access to the Finance share for quarter-end work
Action	Add the user to the group the Finance share allows, then have him sign out and back in
Tools	Active Directory Users and Computers (ADUC), Command Prompt, File Explorer
Outcome	Resolved. After signing back in, the user can open the Finance share

Problem

Bob Nasser (Sales) needs access to the Finance share for quarter-end work. Opening the Finance share as `bnasser` gives a permission error, and `whoami /groups` shows `ACME\GRP-Sales` and **no** `GRP-Finance` (Figure 1).

Background: what an access request is. The user's identity is already proven, since he signs in without trouble. The question is **authorization**: what he is allowed to touch. Granting access to data is the riskiest routine ticket, because the change itself is one click but a careless grant is a security problem. The checks around it matter more than the click.

Validate the request before changing anything

Check	Why
Who is asking?	Verify identity, as in Ticket 01
What exactly is needed?	"Finance" can mean the whole share or one folder
What level of access?	Read-only or edit. Many requests only need read.
Who owns the data?	The data owner (here, the Finance manager) approves, not IT and not the requester's own manager
For how long?	Quarter-end work is temporary. A removal date goes in the ticket.

Sales staff reaching financial records is also a separation-of-duties question, so a missing approval means pausing the ticket and asking, not granting first.

Environment

Item	Detail
Domain	acme.local
Account	bnasser (Bob Nasser), Sales
File server	`\\WIN-4G8SQK9BMC1\Finance` (folder on the server disk: `C:\Shares\Finance`)
Group allowed on the share	`GRP-Finance`
Tools	ADUC, Command Prompt, File Explorer

How access is decided

1. At **sign-in**, the domain controller builds the user's **access token**: his own SID plus the SID of every group he belongs to.

2. When he opens `\\WIN-4G8SQK9BMC1\Finance`, the file server compares that token with the folder's permission lists.

3. Two layers are checked: **share permissions** (the gate at the network door) and **NTFS permissions** (on the folder and files themselves). The effective permission is the more restrictive of the two, and an explicit **Deny** overrides any Allow.

Troubleshooting Steps

#	Action	Result
1	Validate the request (see Problem). Reproduce on the client as `bnasser`: `whoami /groups` and open the Finance share.	Permission error. Only `ACME\GRP-Sales` is listed, no `GRP-Finance`.
2	In ADUC: `Acme / Groups` → `GRP-Finance` → **Members** tab → **Add** → `bnasser` → **Check Names** → OK. Check Names confirms the right account was picked and not a similarly named one.	GRP-Finance members list now includes Bob Nasser.
3	Client: **sign out and back in** (or restart). Locking the screen doesn't count, and `klist purge` doesn't help because it clears Kerberos tickets and not the logon token.	New session builds a new access token.
4	New session: `whoami /groups` and open the Finance share.	Both `ACME\GRP-Finance` and `ACME\GRP-Sales` listed; Finance share opens and shows the test file.
5	Server: check where the files live.	The share is the network entry point to `C:\Shares\Finance`, which contains the test file.

Root Cause

The Finance share allows `GRP-Finance`. Bob wasn't in it, so his token lacked the matching SID and he was refused. Nothing about the share needs to change. **Changing his group membership changes his token.**

Why a group and not his own name on the share: a group answers "who can see Finance?" in one place, is removed in one step when access ends, and doesn't leave forgotten per-user entries behind.

Fix

Add `bnasser` to `GRP-Finance` in ADUC (Troubleshooting Steps, step 2), then have him sign out and back in. The change is in AD immediately, but the user's running session keeps the token it was built with at sign-in.

PowerShell equivalent

```
Add-ADGroupMember -Identity GRP-Finance -Members bnasser
```

Verification

On the new session, `whoami /groups` lists **both** `ACME\GRP-Finance` and `ACME\GRP-Sales`, and the Finance share opens, showing the test file. On the server, the share maps to the folder `C:\Shares\Finance`, which contains the same test file.

Lessons Learned

• **Access level.** The Finance share is set up to give `GRP-Finance` full access at the share level, and the effective permission is the stricter of share and NTFS. Group membership gives Bob whatever that group has, which may be more than the read access quarter-end work needs. If read-only is enough, a separate read-only group is the better answer than adding him here.

• **Temporary access needs an end.** Record the removal date in the ticket. Without one, temporary access quietly becomes permanent. To end it:

```
Remove-ADGroupMember -Identity GRP-Finance -Members bnasser -Confirm:$false
```

• **Audit trail.** Adding a member to a security group is logged on the domain controller as event 4728, which records who made the change, who was added, and to which group:

```
Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4728} -MaxEvents 3 |
  Format-List TimeCreated,Message
```

• **Granting one thing shouldn't open others.** After a change like this, confirm unrelated shares (HR, IT) are still closed to the user.

If it still doesn't work

Symptom	Cause
Still denied after adding the group	The user hasn't signed out and back in (the most common cause)
Added the wrong account	Similar names. Check Names and `Get-ADGroupMember` catch it.
Added to the wrong group	A group that isn't on the share's permission list
Denied on one subfolder only	The NTFS layer. Inspect with `(Get-Acl C:\Shares\Finance).Access`
Can open but not save changes	Read-only was granted, or the NTFS layer is stricter
Works for some users, not others	An explicit Deny entry somewhere. Deny beats Allow.
Delay on a larger network	With several domain controllers, group changes replicate between them

Command reference

```
# Who does the share allow?
Get-SmbShareAccess -Name Finance

# Who is in the group?
Get-ADGroupMember GRP-Finance | Select-Object Name,SamAccountName

# Grant / revoke
Add-ADGroupMember    -Identity GRP-Finance -Members bnasser
Remove-ADGroupMember -Identity GRP-Finance -Members bnasser -Confirm:$false

# In the user's session after signing back in
whoami /groups
```

Screenshots

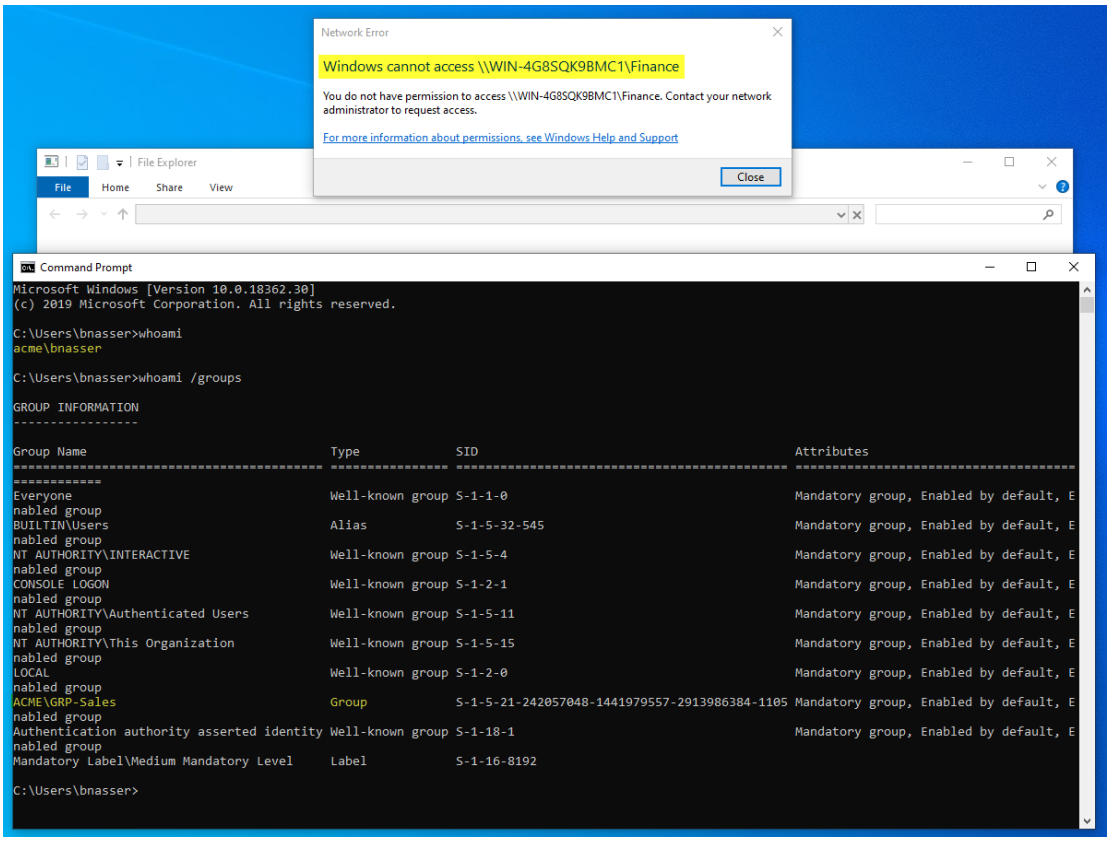


Figure 1. Finance share denied, with whoami /groups showing only GRP-Sales

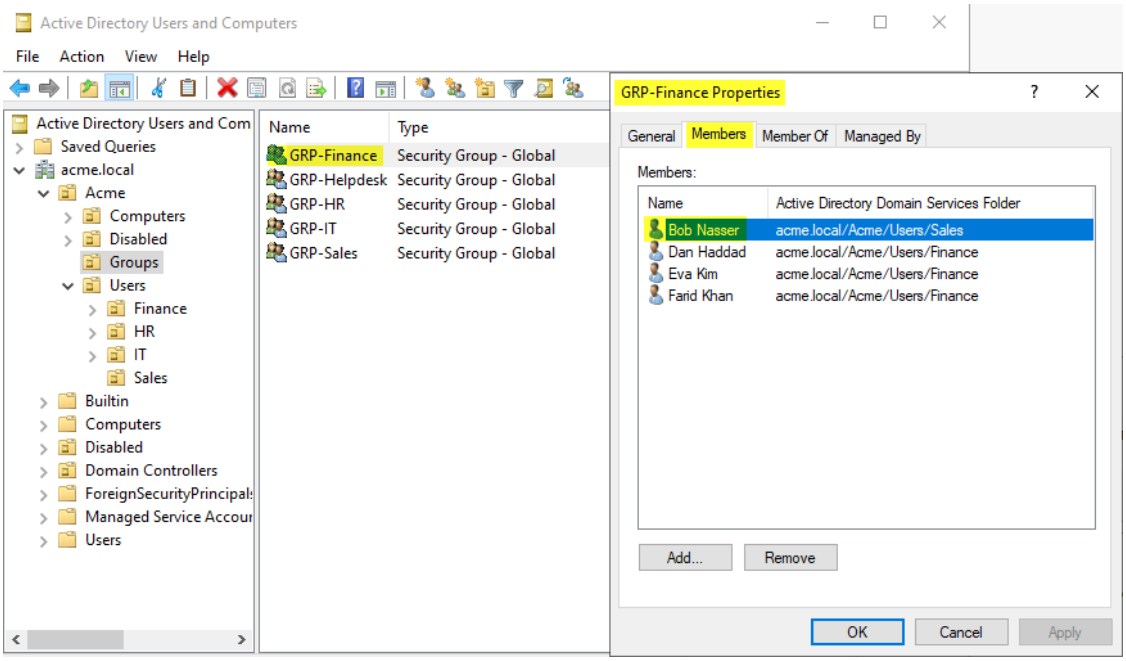


Figure 2. GRP-Finance members list now including Bob Nasser

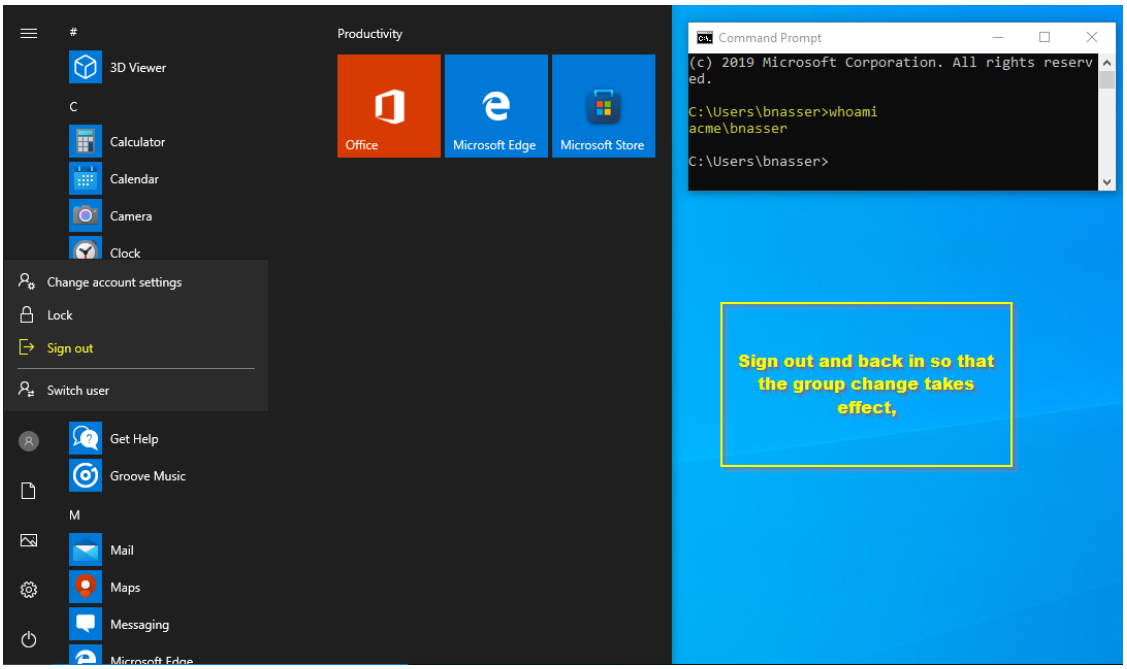


Figure 3. Start menu with Sign out highlighted

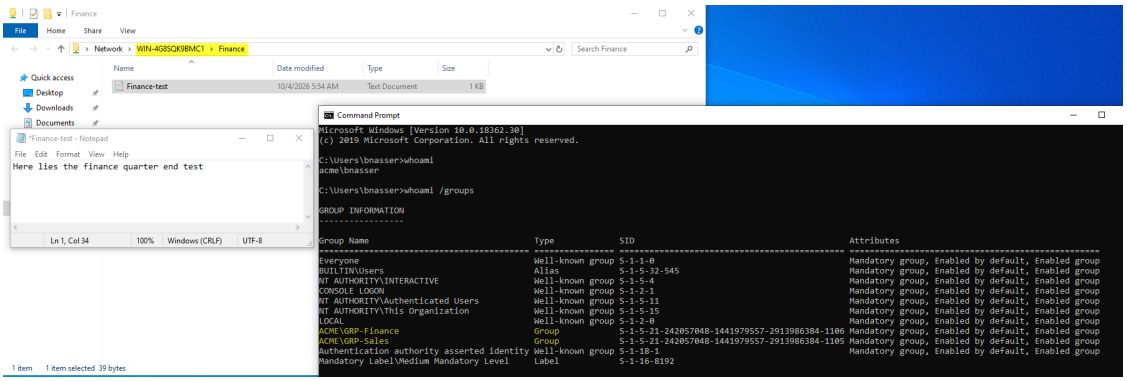


Figure 4. Finance share opened with whoami /groups showing GRP-Finance and GRP-Sales

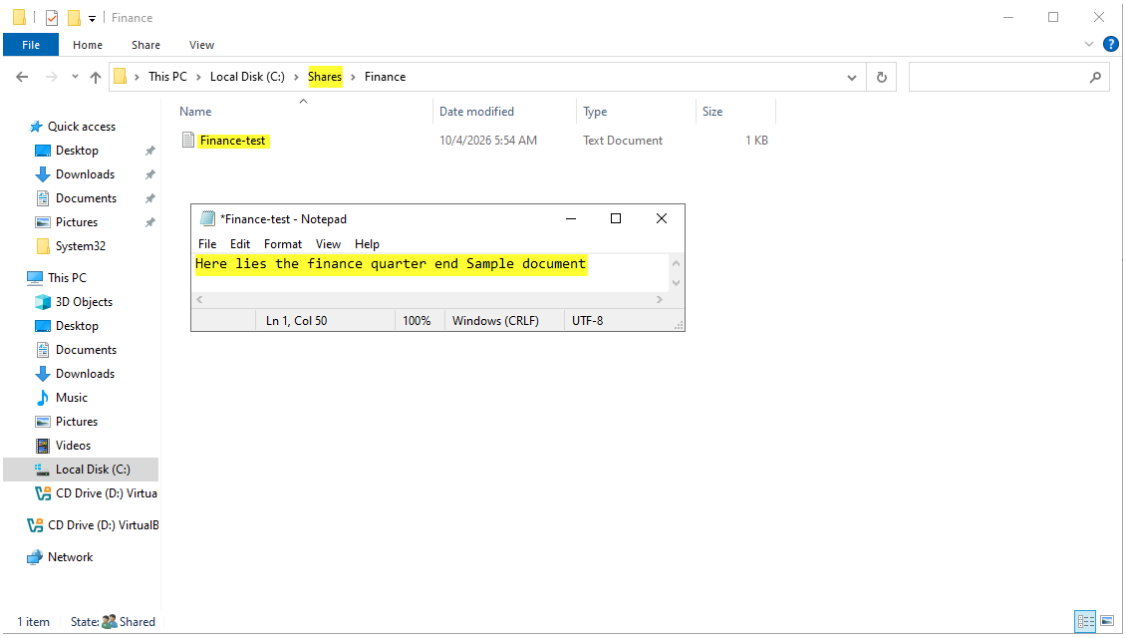


Figure 5. C:\Shares\Finance on the server containing the test file